UNIVERSITY OF SCIENCE, VNU-HCM

Faculty of Information Technology

# CSC14003 - Introduction to Artificial Intelligence

## Project 2: Logic

## Futoshiki Puzzles

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# 1. Introduction

## 1.1. Project Overview

This project implements a complete pipeline for solving *Futoshiki* logic puzzles in the context of CSC14003 — Introduction to Artificial Intelligence. A Futoshiki puzzle is an $N \times N$ grid where each row and column must contain numbers from 1 to $N$ without repetition, while inequality symbols between adjacent cells must also be satisfied.

Our system is organized into clear modules: data I/O, domain modeling, knowledge-base construction, solver implementations, and a GUI for interaction and visualization. The codebase supports loading multiple input instances, running selected solving methods, and exporting results in a reproducible format. This report focuses on two representative approaches: A* search and SAT-based solving.

## 1.2. Objectives and Scope

The main objectives of this project are:

- Model Futoshiki constraints correctly and consistently for both search-based and logic-based solving.
- Implement and evaluate AI solving strategies with readable, modular, and reusable code.
- Build a practical interface for testing puzzle instances of different sizes and difficulties.
- Compare algorithm behavior through solution quality and computational efficiency.

Scope of this report:

- We present the overall project context.
- In the algorithm chapter, we provide detailed descriptions for only two methods: **A* Search** and **SAT Solver (using a SAT library)**.
- Other implemented methods are part of the codebase but are not expanded in detail here.

### 1.3. Team Information

The project was completed by four students from the Faculty of Information Technology, University of Science — VNU-HCM:

- Nguyễn Khánh Toàn — MSSV: 24127252
- Nguyễn Chí Tài — MSSV: 24127529
- Nguyễn Ngọc Thiên — MSSV: 24127545
- Phan Quang Tiến — MSSV: 24127250

The team collaborated on problem modeling, implementation, testing, and report writing, with task assignments summarized in the planning section.

## 2. Project Planning and Task Distribution

### 2.1. Work Breakdown

To ensure traceability to the official grading rubric, the project was decomposed into major work packages that map directly to the required deliverables: FOL modeling, automatic knowledge-base generation, inference algorithms, baseline search algorithms, experimentation, and dissemination artifacts (report and video demonstration). Table 1 summarizes the work structure and expected outputs.

Table 1: Project work breakdown aligned with assignment requirements

| **WP** | **Major Task** | **Expected Deliverable** |
|---|---|---|
| WP1 | FOL formalization for Futoshiki | Complete set of axioms (domain, uniqueness, inequality, clue constraints), plus representative Skolemization and CNF derivations for the report. |
| WP2 | Automatic ground KB/CNF generation | Program module that automatically instantiates puzzle constraints for any grid size $N$ and produces solver-ready clauses/facts. |
| WP3 | Forward Chaining solver | Rule-based inference engine implemented from scratch to derive valid value assignments and detect contradictions. |

| WP | Major Task | Expected Deliverable |
|---|---|---|
| WP4 | Backward Chaining solver | Goal-driven inference mechanism (SLD-style querying) to infer `Val(i,j,v)` facts for unknown cells. |
| WP5 | A* search solver | Heuristic search implementation with admissible state evaluation and constraint-consistency pruning. |
| WP6 | Brute-force and Backtracking baselines | Two baseline solvers used for correctness checking and computational comparison. |
| WP7 | SAT-based solving support | SAT formulation and automated solving pipeline for cross-method verification and performance comparison. |
| WP8 | Experimental setup and I/O pipeline | Input/output standardization, benchmark script, and solved-instance export for all required test cases. |
| WP9 | Technical report and video demo | Final HCMUS-format report (analysis, experiments, discussion) and public demonstration video on representative puzzles. |

## 2.2. Member Responsibilities

Table 2: Member responsibilities

| Member | Primary Responsibilities |
|---|---|
| Nguyễn Khánh Toàn | A* search implementation; SAT-based solver integration; performance export/benchmark pipeline; repository coordination; README and video/demo coordination; GUI integration support; co-editing final report. |
| Nguyễn Chí Tài | Forward Chaining implementation; construction of input test files and puzzle-set preparation; support for report editing and validation tests. |
| Nguyễn Ngọc Thiên | FOL axiom design and CNF transformation workflow; Backward Chaining implementation; support for report editing and consistency checking. |
| Phan Quang Tiến | Automatic KB generation from CNF/grounding logic; Brute-force and Backtracking implementations; support for report editing and comparative analysis sections. |

### 2.3. Completion Percentage per Member

For grading transparency, the team adopted a unified completion declaration at the end of development. As required by this submission, each member is recorded at **100%** completion.

Table 3: Completion percentage per member

| **Member** | **Completion Percentage** |
|---|---|
| Nguyễn Khánh Toàn | 100% |
| Nguyễn Chí Tài | 100% |
| Nguyễn Ngọc Thiên | 100% |
| Phan Quang Tiến | 100% |

### 2.4. Project Management Board

The team managed source code and integration through **GitHub** (branch-based collaboration, pull/merge workflow, and version traceability). Task planning and execution tracking were handled through a shared board (Trello/Planner-like workflow) with status labels (*To do*, *In progress*, *Completed*), priority flags, assignees, and due dates. This process enabled transparent ownership, frequent synchronization, and early detection of integration risks.

Operationally, the management board was synchronized with technical milestones: formalization first, then solver development, then benchmarking and reporting. This ordering reduced late-stage rework and ensured that rubric-weighted items were completed before final packaging.

Table 4: Project timeline and progress board summary

| **Period** | **Stage** | **Main Activities** | **Outputs** | **Status** |
|---|---|---|---|---|
| 08/04/2026–11/04/2026 | Planning and formalization | Requirement analysis, role assignment, FOL specification, initial CNF workflow design. | Work plan, formal FOL draft, initial puzzle inputs. | Completed |
| 12/04/2026–17/04/2026 | Core algorithm implementation | Forward/Backward Chaining, A*, SAT pipeline, KB generation, backtracking baseline modules. | Implemented solver modules and intermediate validation runs. | Completed |
| 18/04/2026–20/04/2026 | Integration and benchmark | Unified I/O handling, solved-output export, runtime comparison, GUI connection and stability checks. | End-to-end executable system and benchmark artifacts. | Completed |
| 21/04/2026–25/04/2026 | Documentation and delivery | Report refinement, figure/table completion, demo video finalization, submission packaging. | Final report PDF, demo link, and complete submission package. | Completed |

# 3. Problem Formulation

## 3.1. Puzzle Rules

Futoshiki is a square-grid logic puzzle played on an $N \times N$ board. The primary objective is to fill the grid with digits such that each row and each column contains every number from 1 to $N$ exactly once. The puzzle is governed by three fundamental categories of constraints:

1. **Latin Square Property:** Similar to Sudoku, every row and every column must be a permutation of the set $\{1, 2, \ldots, N\}$. This implies that no digit can be repeated within the same row or the same column.
2. **Inequality Constraints:** The defining characteristic of Futoshiki is the presence of "less than" ($<$) and "greater than" ($>$) signs between adjacent cells. These signs may be placed horizontally (between cells in the same row) or vertically (between cells in the same column). Any numerical assignment must strictly satisfy these relational constraints.
3. **Pre-filled Clues (Fixed Values):** At the start of the puzzle, some cells may already contain fixed digits (referred to as "Given Clues"). These values are immutable and serve as the starting points for the logical deduction process.

A puzzle is considered solved when every cell is filled with a valid digit and all structural, permutation, and inequality constraints are simultaneously satisfied. For an $N \times N$ board, the total number of cells to be determined is $N^2$, leading to a search space that grows factorially with $N$.

## 3.2. First-Order Logic Formalization

### 3.2.1 Predicate Symbols

The following vocabulary is used to formalize the Futoshiki puzzle rules into First-Order Logic sentences:

- $Val(i, j, v)$: Cell $(i, j)$ is assigned value $v$
- $Given(i, j, v)$: Cell $(i, j)$ has the pre-filled clue value $v$
- $LessH(i, j)$: Horizontal constraint indicating cell $(i, j) < (i, j + 1)$.
- $GreaterH(i, j)$: Horizontal constraint indicating cell $(i, j) > (i, j + 1)$.
- $LessV(i, j)$: Vertical constraint indicating cell $(i, j) < (i + 1, j)$.
- $GreaterV(i, j)$: Vertical constraint indicating cell $(i, j) > (i + 1, j)$.
- $Less(v_1, v_2)$: Integer $v1 < v2$ (background arithmetic relation)

### 3.2.2 Core Axioms

The logic of a Futoshiki puzzle of size $N$x$N$ is encoded using the following universal axioms:

1. **Cell Completeness and Uniqueness:** Each cell must contain exactly one value.

- $\forall i \forall j \exists v\ Val(i, j, v)$
- $\forall i \forall j \forall v_1 \forall v_2 (Val(i, j, v_1) \wedge Val(i, j, v_2) \Rightarrow v_1 = v_2)$

2. **Value Domain and Boundary Constraints:** Values must be within the range $[1, N]$.
   - $\forall i \forall j \forall v (Val(i, j, v) \Rightarrow Less(0, v) \wedge Less(v, N + 1))$
3. **Permutation Constraints (Row and Column Uniqueness):** Every number appears exactly once in each row and column.
   - Row: $\forall i \forall j_1 \forall j_2 \forall v (Val(i, j_1, v) \wedge Val(i, j_2, v) \wedge j_1 \neq j_2 \Rightarrow \perp)$
   - Column: $\forall i_1 \forall i_2 \forall j \forall v (Val(i_1, j, v) \wedge Val(i_2, j, v) \wedge i_1 \neq i_2 \Rightarrow \perp)$
4. **Inequality Constraints:** Satisfaction of all horizontal and vertical relational signs.
   - $\forall i \forall j \forall v_1 \forall v_2 (LessH(i, j) \wedge Val(i, j, v_1) \wedge Val(i, j + 1, v_2) \Rightarrow Less(v_1, v_2))$
   - $\forall i \forall j \forall v_1 \forall v_2 (GreaterH(i, j) \wedge Val(i, j, v_1) \wedge Val(i, j + 1, v_2) \Rightarrow Less(v_2, v_1))$
   - $\forall i \forall j \forall v_1 \forall v_2 (LessV(i, j) \wedge Val(i, j, v_1) \wedge Val(i + 1, j, v_2) \Rightarrow Less(v_1, v_2))$
   - $\forall i \forall j \forall v_1 \forall v_2 (GreaterV(i, j) \wedge Val(i, j, v_1) \wedge Val(i + 1, j, v_2) \Rightarrow Less(v_2, v_1))$
5. **Given Clues Enforcement:** Pre-filled digits must be assigned to their respective cells.
   - $\forall i \forall j \forall v (Given(i, j, v) \Rightarrow Val(i, j, v))$
6. **Background Arithmetic (Ordering Properties)** This axiom defines the strict ordering properties of the *Less* predicate:
   - **Irreflexivity:** $\forall v \neg Less(v, v)$
   - **Transitivity:** $\forall v_1 \forall v_2 \forall v_3 (Less(v_1, v_2) \wedge Less(v_2, v_3) \Rightarrow Less(v_1, v_3))$

   *Note: For practical implementation, the Knowledge Base is populated with ground facts $Less(a, b)$ for all constant pairs $(a, b)$ where $a$ is numerically smaller than $b$ within the domain $[1, N]$.*

### 3.2.3 Step-by-step CNF Derivation

The following examples demonstrate the conversion of FOL axioms into Conjunctive Normal Form (CNF) for use in inference engines.

**Example 1: Cell Value Uniqueness**
FOL Form: $\forall i \forall j \forall v_1 \forall v_2 (Val(i, j, v_1) \wedge Val(i, j, v_2) \Rightarrow v_1 = v_2)$

1. **Step 1 (Eliminate Implication):** Apply the rule $A \Rightarrow B \equiv \neg A \vee B$:

$$\forall i \forall j \forall v_1 \forall v_2 (\neg(Val(i, j, v_1) \wedge Val(i, j, v_2)) \vee (v_1 = v_2))$$

2. **Step 2 (Move Negation Inward):** Apply De Morgan's Law to distribute the negation:

$$\forall i \forall j \forall v_1 \forall v_2 (\neg Val(i, j, v_1) \vee \neg Val(i, j, v_2) \vee (v_1 = v_2))$$

3. **Step 3 (Drop universal quantifiers):** Academic standard

$$\neg Val(i, j, v_1) \vee \neg Val(i, j, v_2) \vee (v_1 = v_2)$$

4. **Step 4 (Standardization for Solver) (Practical):** Since we only consider pairs where $v_1 \neq v_2$ during the grounding process, the literal $(v_1 = v_2)$ evaluates to False ($\perp$) and is removed. **CNF Clause:**

$$[\neg Val(i, j, v_1), \neg Val(i, j, v_2)] \quad \forall v_1 \neq v_2$$

**Example 2: Row Uniqueness (Binary Clauses)**
FOL Form: $\forall i \forall j_1 \forall j_2 \forall v (Val(i, j_1, v) \wedge Val(i, j_2, v) \wedge j_1 \neq j_2 \Rightarrow \perp)$

1. **Step 1 (Eliminate Implication):**

$$\forall i \forall j_1 \forall j_2 \forall v (\neg(Val(i, j_1, v) \wedge Val(i, j_2, v) \wedge j_1 \neq j_2) \vee \perp)$$

2. **Step 2 (Move Negation Inward):** Apply De Morgan's Law and remove the contradiction constant $\perp$:

$$\forall i \forall j_1 \forall j_2 \forall v (\neg Val(i, j_1, v) \vee \neg Val(i, j_2, v) \vee (j_1 = j_2))$$

3. **Step 3 (Drop universal quantifiers):** Academic standard

$$\neg Val(i, j_1, v) \vee \neg Val(i, j_2, v) \vee (j_1 = j_2)$$

4. **Step 4 (Refine Difference Constraint) (Practical):** Since the axiom iterates over distinct column indices, this literal is False and is

discarded during grounding. **CNF Clause:**

$$[\neg Val(i, j_1, v), \neg Val(i, j_2, v)] \quad \forall j_1 \neq j_2$$

**Example 3: Horizontal Less-than**
FOL Form: $\forall i \forall j \forall v_1 \forall v_2 (LessH(i, j) \wedge Val(i, j, v_1) \wedge Val(i, j + 1, v_2) \Rightarrow Less(v_1, v_2))$

1. **Step 1 (Eliminate Implication):**

$$\forall i \forall j \forall v_1 \forall v_2 (\neg(LessH(i, j) \wedge Val(i, j, v_1) \wedge Val(i, j+1, v_2)) \vee Less(v_1, v_2))$$

2. **Step 2 (Move Negation Inward):** Apply De Morgan's Law to distribute the negation across the conjunction:

$$\forall i \forall j \forall v_1 \forall v_2 (\neg LessH(i, j) \vee \neg Val(i, j, v_1) \vee \neg Val(i, j+1, v_2) \vee Less(v_1, v_2))$$

3. **Step 3 (Drop Universal Quantifiers):** Remove the quantifiers to obtain the final clausal form: **CNF Clause:**

$$[\neg LessH(i, j), \neg Val(i, j, v_1), \neg Val(i, j + 1, v_2), Less(v_1, v_2)]$$

## 3.3. Forward Chaining (Maintaining Arc Consistency)

### 3.3.1 Idea and Architecture

**Why MAC instead of pure Forward Chaining or DPLL**
Pure Forward Chaining (FC) starts from known facts and applies Modus Ponens exhaustively until no new fact can be derived. For Futoshiki puzzles with few given clues, FC alone terminates with an *incomplete* assignment: many cells still have multiple candidate values and no rule fires to resolve them. The solver is stuck without guessing.

DPLL (Davis–Putnam–Logemann–Loveland), on the other hand, operates on a *propositional CNF* encoding. Every cell-value pair becomes a Boolean variable; the KB is a flat list of clauses. Unit Propagation plays the role of Forward Chaining at the Boolean level. While correct, the CNF representation discards the structure of the problem: a "domain" is implicit across $N$ Boolean variables per cell and must be reconstructed during

branching.

**MAC (Maintaining Arc Consistency)** retains the native *domain* representation — each cell $(i, j)$ holds a set of candidate values $D_{i,j} \subseteq \{1, \ldots, N\}$ — and interleaves FC-style propagation with backtracking search. Every time the search makes a tentative assignment, FC is re-run on the updated domain to prune inconsistent values from neighbouring cells before recursing deeper. This *maintains* arc consistency throughout the search, detecting dead ends far earlier than either pure FC or DPLL.

Concretely, the implementation is organised in two tiers:

- **Inference engine (`_run_fc`):** A FOL Forward Chaining engine that processes an agenda of typed facts $\{\texttt{Assign}(i, j, v), \texttt{Remove}(i, j, v)\}$ and applies five Modus Ponens rules (described below) until the agenda is empty or a contradiction is detected.
- **Search controller (`_backtrack`):** When FC cannot reduce any domain further, a branching step selects the most constrained unassigned cell (MRV), guesses a value, invokes FC, and recurses. On failure, the domain snapshot is restored and the next value is tried.

The key insight is that `_run_fc` is called *both* as an initial propagation pass over the puzzle clues and as the consistency- maintenance step after every branching decision — the same code plays both roles, which is the defining characteristic of MAC.

### 3.3.2 Algorithm Analysis

**Domain Representation**

Each cell $(i, j)$ maintains a set $D_{i,j} \subseteq \{1, \ldots, N\}$. A cell is considered *assigned* when $|D_{i,j}| = 1$. The initial state sets $D_{i,j} = \{1, \ldots, N\}$ for every cell.

**Initial Facts**

The puzzle instance is converted into a seed agenda of typed facts:

- $\text{Given}(i, j, v) \Rightarrow \texttt{Assign}(i, j, v)$
- $\text{LessH}(i, j): \ (i, j) < (i, j{+}1) \Rightarrow \texttt{Remove}(i, j, N)$ and $\texttt{Remove}(i, j{+}1, 1)$
- $\text{GreaterH}(i, j): \ (i, j) > (i, j{+}1) \Rightarrow \texttt{Remove}(i, j, 1)$ and $\texttt{Remove}(i, j{+}1, N)$
- Similarly for vertical constraints LessV and GreaterV.

These correspond directly to the ground instances of FOL axioms A4–A9.

**Five Modus Ponens Rules (FC Engine)**

The FC engine applies the following rules on each fact popped from the agenda:

| Rule | Trigger | Effect |
|---|---|---|
| R1 | `Assign`$(i, j, v)$ arrived, $v \notin D_{i,j}$ | Contradiction $\Rightarrow$ return *False* |
| R2 | `Remove`$(i, j, v)$ makes $D_{i,j} = \emptyset$ | Contradiction $\Rightarrow$ return *False* |
| R3 | `Assign`$(i, j, v)$ | Push `Remove`$(i, c, v)$ for all $c \neq j$ (row)<br>Push `Remove`$(r, j, v)$ for all $r \neq i$ (column) |
| R4 | $\lvert D_{i,j}\rvert = 1$ after a Remove | Push `Assign`$(i, j, v')$ where $v'$ is the sole survivor |
| R5 | Any domain change at $(i, j)$ | Propagate new min / max of $D_{i,j}$ to all four inequality |

Rules R1–R4 correspond to the grounded FOL axioms A1–A3 (cell uniqueness and row/column permutation). Rule R5 implements A4–A9 (inequality constraints): if $(i, j) < (i, j{+}1)$, then every value $v \leq \min(D_{i,j})$ is infeasible for $(i, j{+}1)$ and is pushed as a `Remove` fact.

**Backtracking with MRV**

When the FC engine returns without fully assigning all cells, the search controller:

1. Selects the unassigned cell with the *minimum remaining values* (MRV) in its domain — the fail-first heuristic.
2. Takes a snapshot of the entire domain matrix.
3. Tries each value $v \in D_{i,j}$ in ascending order. For each attempt, it pushes `Assign`$(i, j, v)$ onto a fresh agenda and calls `_run_fc`.
4. On success, recurses. On failure (FC returns *False* or the recursive call fails), restores the snapshot and tries the next $v$.

**Complexity Analysis**

| Aspect | Bound | Remark |
|---|---|---|
| Domain size | $O(N^3)$ | $N^2$ cells, each domain $\leq N$ |
| FC per node | $O(N^3)$ | Each value removed at most once |
| MRV selection | $O(N^2)$ | Scan all cells; early exit at size 2 |
| Max search depth | $\leq N^2$ | One cell assigned per level |
| Backtracks (worst) | $O(N!)$ | Full combinatorial space |
| Backtracks (typical) | $O(N^2)$ | MRV heuristic combined with arc consistency eliminates combinatorial explosion on large inputs |
| Space per snapshot | $O(N^3)$ | Copy of all domain sets |

**Advantages and Limitations**

- **Stronger pruning than DPLL:** MAC propagates inequality constraints through min/max bounds on domain sets, a finer-grained operation than Boolean unit propagation. Infeasible branches are detected several levels earlier. This lightweight micro-operation allows MAC to process tens of thousands of nodes in fractions of a second, as detailed in the Experiments section.
- **Directly mirrors FOL semantics:** Facts and rules are typed (Assign / Remove) and match the FOL predicates $Val(i, j, v)$, LessH, GreaterH, etc. without a translation layer.
- **Complete:** Backtracking guarantees a solution is found whenever one exists.
- **No clause learning:** Unlike CDCL-based SAT solvers, this implementation does not record conflict clauses. Repeated conflicts in structurally similar sub-trees are re-explored.
- **Snapshot cost:** Each backtracking node copies the full $O(N^3)$ domain matrix. For $N = 9$ this is manageable ($\approx 729$ integers) but grows rapidly with $N$.

### 3.3.3 Pseudocode

**Algorithm 1** MAC Master Coordinator

```
function SOLVE(puzzle)
    D ← initialise all domains to {1, ..., N}
    facts_0 ← InitialFacts(puzzle)          ▷ Given clues + boundary inequalities
    if not RunFC(D, facts_0) then
        return None                         ▷ Contradiction in initial facts
    end if
    if IsSolved(D) then
        return ExtractGrid(D)
    end if
    if Backtrack(D) then
        return ExtractGrid(D)
    end if
    return None
end function
```

**Algorithm 2** FC Engine — Modus Ponens on Domain Facts

```
function RunFC(D, facts)
    agenda ← DEQUE(facts);  seen ← {facts};  assigned ← ∅
    while agenda is not empty do
        fact ← agenda.POPLEFT
        if fact = Assign(i, j, v) then
            if v ∉ D[i][j] then return False                ▷ R1: contradiction
            end if
            if (i, j) ∉ assigned then
                D[i][j] ← {v};  assigned.ADD((i, j))
                for c ≠ j do PUSH(Remove(i, c, v))
                end for                                     ▷ R3: row
                for r ≠ i do PUSH(Remove(r, j, v))
                end for                                     ▷ R3: column
                PropagateIneq(i, j, D, Push)                ▷ R5
            end if
        else if fact = Remove(i, j, v) then
            if v ∉ D[i][j] then continue
            end if
            D[i][j].DISCARD(v)
            if D[i][j] = ∅ then return False                ▷ R2: contradiction
            end if
            if |D[i][j]| = 1 then
                PUSH(Assign(i, j, sole value of D[i][j]))   ▷ R4
            end if
            PropagateIneq(i, j, D, Push)                    ▷ R5
        end if
    end while
    return True
end function
```

**Algorithm 3** Inequality Propagation (Rule R5)

```
function PropagateIneq(i, j, D, push)
    lo ← min(D[i][j]);  hi ← max(D[i][j])
    for each neighbour (i′, j′) of (i, j) connected by an inequality do
        if (i, j) < (i′, j′) then                    ▷ (i, j) is the smaller side
            for v ∈ D[i′][j′] with v ≤ lo do
                push(Remove(i′, j′, v))
            end for
        else if (i, j) > (i′, j′) then               ▷ (i, j) is the larger side
            for v ∈ D[i′][j′] with v ≥ hi do
                push(Remove(i′, j′, v))
            end for
        end if
    end for
end function
```

**Algorithm 4** MAC Backtracking with MRV

```
function Backtrack(D)
    (i, j) ← MRV(D)        ▷ Cell with smallest |D[i][j]| > 1; None if all assigned
    if (i, j) = None then return True
    end if
    for v ∈ sorted(D[i][j]) do
        D_saved ← DeepCopy(D)
        if RunFC(D, [Assign(i, j, v)]) then
            if Backtrack(D) then return True
            end if
        end if
        D ← D_saved                                  ▷ Restore snapshot
    end for
    return False
end function
```

### 3.3.4 Illustration

Consider a $4 \times 4$ Futoshiki puzzle with the following initial state: cell $(0, 0)$ is given as 2, cell $(3, 3)$ is given as 4, and there is a single inequality constraint $(0, 1) > (0, 2)$. The initial agenda is populated with these base facts.

1. **FC pass 1 (Initial Propagation):** The `Assign`$(0, 0, 2)$ and `Assign`$(3, 3, 4)$ facts are processed. Rule R3 pushes `Remove` facts to eliminate 2 from row 0 and column 0, and 4 from row 3 and column 3. The inequality $(0, 1) > (0, 2)$ triggers Rule R5, pushing `Remove`$(0, 1, 1)$ (as the larger cell must be strictly greater than the absolute minimum 1) and `Remove`$(0, 2, 4)$ (as the smaller cell must be strictly less than the absolute maximum 4). As the domain $D[0, 1]$ loses values 1 (due to the inequality) and 2 (due to row 0), its remaining domain becomes $\{3, 4\}$. The domain $D[0, 2]$ loses 2 and 4, narrowing down to $\{1, 3\}$.
2. **FC Stuck:** The agenda eventually becomes empty. Several cells still have $|D| > 1$. The `IsSolved` function evaluates to *False*, meaning the pure deduction phase has halted without a complete assignment.
3. **Backtracking (MRV) and Cascade:** The search controller selects the unassigned cell with the Minimum Remaining Values. Suppose it picks $(0, 1)$ with $D = \{3, 4\}$. It first guesses $v = 3$ by pushing `Assign`$(0, 1, 3)$ into a fresh agenda and invoking `RunFC`. This triggers Rule R3, pushing `Remove`$(0, 2, 3)$ to eliminate 3 from the rest of row 0. The domain $D[0, 2]$ (which was previously $\{1, 3\}$) loses 3, reducing its size to exactly 1 ($D[0, 2] = \{1\}$). Rule R4 immediately fires, pushing `Assign`$(0, 2, 1)$. Concurrently, Rule R5 propagates the new $\max(D[0, 1]) = 3$, enforcing that $(0, 2)$ must be $< 3$, which perfectly aligns with the new singleton domain. This single heuristic guess creates a powerful chain reaction (cascade) that rapidly collapses the remaining domains, successfully finding the solution without needing further backtracks.

The total number of backtracks on this instance is typically zero or one, because arc consistency eliminates almost all ambiguity after the first FC pass.

## 3.4. Backward Chaining (SLD Resolution)

### 3.4.1 Idea

The backward-chaining solver developed by our team operates as a hybrid of Prolog-style SLD resolution and backtracking search.

- **KB construction($HornConverter$):** Converts a Futoshiki instance into a Knowledge Base (KB) consisting of facts and definite Horn rules over predicates like $val(i, j, v)$ and $not_val(i, j, v)$.
- **Backward chaining engine ($fol_bc_or$ / $fol_bc_and$):** Prolog-like interpreter implemented from scratch that answers queries using unification and depth-first goal reduction.
- **Search controller ($_backtrack$):** Responsible for filling the grid. For each empty cell $(r, c)$, the controller calls the backward chaining engine to enumerate valid candidates for $val(r, c, v)$. It then asserts the chosen val as a new fact in the KB, recurses to the next cell, and retracts the fact upon failure.

**Why search is integrated with SLD Resolution:** The fundamental First-Order Logic (FOL) constraint for Futoshiki states that "every cell has at least one value" ($\forall i \forall j \exists v\ Val(i, j, v)$). This existential quantification is not expressible as a definite Horn rule. Therefore, a pure Prolog-style engine cannot natively generate assignments without complex, inefficient workarounds. To resolve this, we employ choice + backtracking in Python to build a full assignment incrementally, while utilizing the Horn rules within the BC engine purely to derive exclusions ($not_val$) and prune invalid branches.

**Technical Implementation Notes:**

- **0-based Indexing:** While the formal FOL axioms in this report use 1-based indexing (as per the mathematical specification $1..N$), the internal engine and the trace examples below utilize 0-based indexing. This engineering decision aligns naturally with Python's List data structures, eliminating redundant $+1/-1$ arithmetic at every node expansion and optimizing low-level computational speed.
- **Hybrid vs. Pure Prolog:** A pure Prolog implementation solving a $7 \times 7$ or higher grid via a single query would generate an SLD tree so deep it would cause a RecursionError or a memory blowout in Python. The Hybrid architecture ensures the memory footprint remains manageable by using BC as a highly precise "Domain Pruning Oracle" for individual cells, rather than a global solver.
- **Partial Grounding:** To mitigate combinatorial explosion during

unification, our implementation uses Partial Grounding. Static constraints (like boundary inequalities) are fully grounded into facts (e.g., $not_val(0, 1, 3)$), while the core generator rules are kept in First-Order form with variables (e.g., $val(i, j, v)$). This drastically reduces the size of the KB while retaining the dynamic power of SLD resolution.

### 3.4.2 Algorithm Analysis

**Atoms (Predicates)**

The Horn KB used the following vocabulary:

- $val(i, j, v)$: Cell $(i, j)$ is assigned the value $v$.
- $given(i, j, v)$: Cell $(i, j)$ has the pre-filled clue value $v$
- $less_h(i, j)$ / $greater_h(i, j)$: Horizontal inequalities between $(i, j)$ and $(i, j + 1)$.
- $less_v(i, j)$ / $greater_v(i, j)$: Vertical inequalities between $(i, j)$ and $(i + 1, j)$.
- $less(v1, v2)$: Background arithmetic fact representing $v_1 < v_2$.
- $neq(x, y)$: Background inequality fact representing $x \neq y$.
- $not_val(i, j, v)$: Value $v$ is explicitly forbidden at cell $(i, j)$.

**Facts**

- **Background facts**: All valid $less(v1, v2)$ and $neq(x, y)$ combinations within the finite domain $1..N$.
- **Instance facts**: $given(...)$ clues, inequality signs, and immediate "boundary pruning" facts ($not_val(...)$) derived directly from extreme inequality conditions (e.g., a cell on the smaller side of an inequality cannot hold the maximum value $N$).

**Rules (Definite Horn)**

Constraints are strictly encoded as rules deriving $not_val$:

- Given $\Rightarrow Val : val(i, j, v) : -given(i, j, v)$
- Cell uniqueness: If a value is chosen, all others are forbidden in that cell.
- Row/Col uniqueness: If $val(i, j1, v)$ holds, then $not_val(i, j2, v)$ for all $j_2 \neq j_1$.

- Inequalities: Values that violate a $<$ or $>$ constraint trigger a $not_val$ derivation.

**Existence / Choice Rule via NAF**

To generate valid candidates for an empty cell, the following rule is injected into the KB:$val(i, j, v) : -domain(v), naf(not_val(i, j, v))$ Explain: A value $v$ is a valid candidate for cell $(i, j)$ IF it exists in the domain AND the system fails to prove that it is forbidden ($not_val$). This utilizes "Negation as Failure" (NAF), enabling non-monotonic reasoning over our finite Knowledge Base.

**Unification and Substitution**

- $unify(x, y, \theta)$: Computes the Most General Unifier (MGU) for two expressions, returning the substitution dictionary $\theta$ or "failure".
- $subst(\theta, expr)$: Applies the substitution $\theta$ to goals and rule bodies. Variables are represented as lowercase strings (e.g., 'v', 'i', 'j').

**SLD Resolution (AND/OR Search Tree)**

- OR node ($bc_or$): Proves a goal by attempting to match and unify it against the head of any available fact or rule.
- AND node ($bc_and$): Proves the body of a rule by proving all subgoals sequentially from left to right.
- This mimics the standard Prolog evaluation strategy: depth-first, leftmost.

**Negation as Failure (NAF) Handling**

- naf(G) succeeds if and only if the engine exhausts all paths and cannot derive G.
- Safety Implementation: When evaluating inside a NAF block ($in_naf = True$), the engine restricts derivations for val to facts only, ignoring rules. This prevents fatal self-supporting cycles (e.g., attempting to prove val by assuming not $not_val$, which loops back to val), ensuring termination and soundness.

**Optimizations Implemented**

- Fact and Rule Indexing: Knowledge is partitioned using dictionaries $(fact_index[pred], pred_rule_index, exact_rule_index)$ to ensure $O(1)$ lookup times instead of scanning the entire KB sequentially.Per-query
- Memoization: Subgoal results are cached (self.memo) during a single $bc_ask$ lifecycle to prevent recomputation of identical states.
- Cycle Detection (visited set): Tracks the active resolution path to prevent infinite recursion in cyclic goal graphs.
- Boundary Pruning: Converting extreme inequalities into immediate $not_val$ facts drastically reduces the search space before resolution even begins.

### 3.4.3 Pseudocode

**Algorithm 5** Master Coordinator

```
function SOLVE(puzzle)
    kb ← HornConverter(puzzle).build()
    add facts domain(1..N)
    add facts val(...)
    for givens in the grid do
        add rule schema: val(i, j, v) : −domain(v), naf(not_val(i, j, v))
        build indexes for facts/rules
    end for

    for each given (r, c) = v do
        if FOL_BC_ASK(kb, not_val(r, c, v) succeeds then
            return failure // Contradiction in givens
        end if
        return _BACKTRACK(kb)
    end for
end function
```

**Algorithm 6** State Space Manager

```
function BACKTRACK(kb)
    if grid complete then
        return success
    end if
    (r, c) ← first empty cell
    for each answer substitution theta in FOL_BC_ASK(kb, val(r, c, v)) do
        value ← theta[v]
        grid[r][c] ← value
        assert fact val(r, c, value) into kb
        if _BACKTRACK(kb) success then
            return success
        end if
        retract fact val(r, c, value) from kb
        grid[r][c] ← 0
    end for
    return failure
end function
```

**Algorithm 7** Inference Interface

```
function FOL_BC_ASK(kb, goal)
    memo ← ∅                                        ▷ Init Memoization
    return FOL_BC_OR(kb, goal, theta = ∅, in_naf = false, visited = ∅)
end function
```

**Algorithm 8** Goal Resolver - OR Node

```
function FOL_BC_OR(kb, goal, theta, in_naf, visited)
    g ← SUBST(θ, goal)
    if (g, in_naf) ∈ visited then
        return ∅                                        ▷ Cycle detection
    end if
    visited′ ← visited ∪ {(g, in_naf)}
    if g is naf(inner) then                             ▷ Negation As Failure
        if FOL_BC_OR(kb, inner, theta, true, visited′) is None then
            return {theta}                              ▷ NAF success
        else
            return ∅                                    ▷ NAF failure
        end if
    end if
    answers ← ∅
    for each fact F matching g in kb do
        theta′ ← UNIFY(args(g), args(head(F)), theta)
        if theta′ ≠ failure then
            answers ← answers∪FOL_BC_AND(kb, body(F), theta′, in_naf, visited′)
        end if
    end for
    if not (in_naf and pred(g) == val) then
        for each rule R matching g in kb do
            theta′ ← UNIFY(args(g), args(head(R)), theta)
            if theta′ ≠ failure then
                answers ← answers∪FOL_BC_AND(kb, body(R), theta′, in_naf, visited′)
            end if
        end for
    end if
    return answers
end function
```

**Algorithm 9** Rule Body Validator - AND Node

```
function FOL_BC_AND(kb, goals, theta, in_naf, visited)
    if goals is None then
        return {theta}                ▷ Chứng minh thành công toàn bộ Body
    end if
    first ← SUBST(theta, goals[0])
    rest ← goals[1 :]                 ▷ Các mục tiêu còn lại
    answers ← ∅
    for each theta′ in FOL_BC_OR(kb, first, theta, in_naf, visited) do
        answers ← answers∪FOL_BC_AND(kb, rest, theta′, in_naf, visited)
    end for
    return answers
end function
```

### 3.4.4 Illustration

[Chèn nội dung tại đây]

## 3.5. A* Search

### 3.5.1 Idea

In this project, each search state is a partial Futoshiki assignment, where value 0 denotes an unfilled cell. The A* solver expands states according to

$$f(s) = g(s) + h(s),$$

with the following cost model:

- Each action assigns exactly one value to one empty cell, so each step has unit cost.
- $g(s)$ is the number of assignments already made from the initial state.
- $h(s)$ estimates the minimum number of assignments still needed to reach a complete solution.

The transition model assigns one value to one unfilled cell. To reduce branching, the solver selects the next variable by MRV (Minimum Remaining Values), i.e., the cell with the smallest candidate domain. Before evaluating child states, domains are filtered by row/column uniqueness and inequality constraints; optional AC-3 propagation is then applied to remove unsupported values.

This design combines informed search (A*) with CSP-style filtering (MRV + AC-3), making the exploration significantly more directed than naive uninformed expansion.

### 3.5.2 Heuristic Function and Admissibility

Let $U(s)$ be the number of unassigned cells in state $s$. We choose:

$$h(s) = U(s).$$

Equivalently, if $N$ is board size and $A(s)$ is number of assigned cells, then

$$h(s) = N^2 - A(s).$$

Feasibility pruning with AC-3 is used as a separate consistency check: if propagation yields an empty domain, the state is discarded as inconsistent.

**Why this heuristic is admissible:**

- From a partial state with $U(s)$ empty cells, any complete solution must assign all those $U(s)$ cells.
- Each action can assign at most one cell.
- Therefore the true remaining cost $h^*(s)$ satisfies $h^*(s) \geq U(s) = h(s)$.

Hence $h(s)$ never overestimates the optimal remaining cost, so it is admissible.

**Consistency (monotonicity):**

- After one valid assignment $s \to s'$, we have $U(s') = U(s) - 1$.
- With unit step cost $c(s, s') = 1$:

$$h(s) = U(s) = 1 + U(s') = c(s, s') + h(s').$$

- So $h$ is consistent as well.

### 3.5.3 Pseudocode

**Algorithm 10** A* for Futoshiki (with MRV and AC-3)

```
Build initial state s_0 from puzzle
Validate givens; if invalid then return failure
Compute domains and heuristic for s_0
Initialize closed as an empty set
Initialize best_g[s_0]
Push s_0 into priority queue by key (f, h, −g)
while priority queue not empty do
    Pop best state s
    if s ∈ closed then
        continue
    end if
    Add s to closed
    if s is goal then return solved grid
    end if
    Select unassigned cell (r, c) by MRV
    for each value v in domain(r, c) do
        if assigning v is consistent then
            Create child state s′
            if s′ ∈ closed then
                continue
            end if
            Compute g(s′)
            if best_g[s′] exists and g(s′) is not better then
                continue
            end if
            Recompute domains (and AC-3 if enabled)
            if no empty domain in s′ then
                Set g(s′) = g(s) + 1, evaluate h(s′), f(s′) = g(s′) + h(s′)
                Update best_g[s′]
                Push s′ into priority queue
            end if
        end if
    end for
end while
return failure
```

### 3.5.4 Illustration

Consider a $4 \times 4$ instance. After loading givens and inequalities, the solver:

1. Initializes domains for all empty cells.
2. Runs consistency filtering; if AC-3 is enabled, unsupported values are removed transitively.
3. Chooses the most constrained empty cell by MRV.
4. Expands candidate assignments in priority order according to $f = g+h$.

Example progression:

- State $s_0$: 4 cells fixed from clues, several constrained inequalities.
- State $s_1$: assign value to the MRV cell (domain size 1 or 2), causing further domain reduction.
- State $s_2$: another forced assignment appears after propagation.
- Goal state: all cells filled, row/column uniqueness and inequalities all satisfied.

This illustrates how propagation shrinks the search tree before deep branching happens.

### 3.5.5 Algorithm Analysis

Let $N$ be puzzle size (grid $N^2$ cells), and let $b$ be average branching factor after filtering.

**Time complexity (worst case):**

- Search may still be exponential in the number of unassigned cells, roughly $O(b^d)$ with depth $d \leq N^2$.
- AC-3 propagation on each evaluated state adds polynomial overhead; practical runtime is dominated by number of expanded states.

**Space complexity:**

- A* stores open/closed states and domain data, so memory usage is higher than depth-first methods.
- In worst case, space is also exponential in search depth.

**Practical behavior:**

- The heuristic $h(s) = U(s)$ is admissible and easy to justify, which is suitable for the project requirement on heuristic correctness.
- Because this heuristic is simple, most practical speedup comes from MRV ordering and AC-3 pruning.

- The *closed* set avoids re-expanding visited states, and *best_g* avoids keeping inferior paths to the same state.
- The method is robust for medium-sized puzzles, but memory pressure can increase for larger or weakly constrained instances.

## 3.6. Brute-force and Backtracking

### 3.6.1 Idea

Brute-force is a straightforward approach that tries all possible assignments for the empty cells in the Futoshiki puzzle. For a puzzle of size $N \times N$, each empty cell can take a value from 1 to $N$. The algorithm generates all possible combinations and checks whether they satisfy the constraints (row, column, and inequality constraints). However, this approach is highly inefficient due to the exponential number of possibilities.

Backtracking is an optimized version of brute-force. Instead of generating all possible assignments, it builds the solution step by step and abandons a partial assignment as soon as it violates any constraint. At each step, the algorithm selects an empty cell, tries a valid value, and recursively proceeds to the next cell. If a conflict occurs, it backtracks and tries another value. This significantly reduces the search space compared to brute-force.

### 3.6.2 Pseudocode

**Brute-force:**

**Algorithm 11** Brute-force

```
function BRUTE_FORCE(grid):
    empty_cells = list of all empty positions
    for each possible assignment of values to empty_cells:
        fill grid with assignment
        if is_valid(grid):
            return grid
    return failure
```

**Backtracking:**

**Algorithm 12** Backtracking

```
function BACKTRACK(grid):
    if no empty cell:
        return grid

    (row, col) = find an empty cell

    for value from 1 to N:
        if is_valid(row, col, value):
            grid[row][col] = value

            result = BACKTRACK(grid)
            if result != failure:
                return result

            grid[row][col] = empty

    return failure
```

### 3.6.3 Illustration

Consider a $4 \times 4$ Futoshiki puzzle with some empty cells.

Brute-force will generate all possible combinations for the empty cells. For example, if there are $k$ empty cells, it will try up to $N^k$ assignments, checking each one against all constraints.

Backtracking, on the other hand, fills the grid step by step. Suppose it assigns a value that violates a constraint (e.g., duplicates in a row or breaks an inequality). Instead of continuing, it immediately backtracks and tries another value. This pruning avoids exploring invalid branches of the search tree, making it much more efficient in practice.

### 3.6.4 Algorithm Analysis

**Brute-force Complexity:** The time complexity is $O(N^k)$, where $k$ is the number of empty cells. In the worst case (empty grid), this becomes $O(N^{N^2})$, which is infeasible for large $N$.

**Backtracking Complexity:** The worst-case time complexity is still exponential, but significantly reduced in practice due to pruning. The actual

performance depends on how early invalid assignments are detected.

**Space Complexity:** Backtracking uses recursion, so the space complexity is $O(N^2)$ for the grid plus $O(N^2)$ for the recursion stack in the worst case.

**Comparison:**

- Brute-force explores all possibilities without pruning.
- Backtracking prunes invalid states early, making it much more efficient.
- Backtracking is the preferred method for solving constraint satisfaction problems like Futoshiki.

## 3.7. SAT Solver (Using SAT Library)

### 3.7.1 Idea

The SAT approach transforms the Futoshiki puzzle into a propositional satisfiability problem in CNF (Conjunctive Normal Form). Instead of searching directly over puzzle states, we encode all puzzle rules as logical clauses and let a SAT engine decide whether there exists a satisfying assignment.

In this project, CNF clauses are generated by the `FOLKB` module, then solved by PySAT (Glucose3). If SAT, the returned model is decoded back to an $N \times N$ grid; if UNSAT, the puzzle has no valid solution under given constraints.

### 3.7.2 Constraint Encoding

For each triple $(i, j, v)$, define boolean variable $X_{i,j,v}$:

$$X_{i,j,v} = \text{true} \iff \text{cell } (i, j) \text{ has value } v.$$

Variable ID mapping used in code:

$$\text{ID}(i, j, v) = i \cdot N^2 + j \cdot N + v,$$

with $i, j \in [0, N-1]$, $v \in [1, N]$, so IDs are in $[1, N^3]$.

Main CNF groups:

- **Cell at least one value:** each cell must take some $v \in \{1, \ldots, N\}$.

- **Cell at most one value:** no cell can take two different values simultaneously.
- **Row uniqueness:** two different columns in same row cannot share the same value.
- **Column uniqueness:** two different rows in same column cannot share the same value.
- **Given clues:** pre-filled cells are encoded as unit clauses.
- **Inequality constraints:** for each adjacent pair with sign $<$ or $>$, all violating value-pairs are forbidden by binary clauses.

This encoding guarantees that any satisfying model corresponds to a valid Futoshiki solution.

### 3.7.3 Pseudocode

**Algorithm 13** SAT-based Futoshiki Solver

```
Build CNF clauses from puzzle constraints using FOLKB
Create SAT solver instance (Glucose3)
Add all CNF clauses to solver
if solver returns UNSAT then
    return failure
end if
Get SAT model (set of literals)
Initialize empty grid N × N
for each positive literal ℓ in model do
    if 1 ≤ ℓ ≤ N^3 then
        Decode ℓ → (i, j, v)
        Set grid(i, j) ← v
    end if
end for
if any cell remains unassigned then
    return failure
else
    return solved grid
end if
```

### 3.7.4 Illustration

For a $4 \times 4$ puzzle:

1. The encoder creates variables $X_{i,j,v}$ for all $4^3 = 64$ combinations.
2. Clauses are generated for Latin constraints (row/column uniqueness), given clues, and inequality symbols.
3. Glucose3 finds a satisfying model, for example including literals that decode to assignments such as $(0, 0) = 2$, $(0, 1) = 4$, etc.
4. After decoding all positive in-range literals, the grid is fully reconstructed and validated as complete.

Compared with incremental state expansion, the SAT method delegates global consistency reasoning to a highly optimized conflict-driven SAT engine.

#### 3.7.5 Algorithm Analysis

**Encoding cost:**

- Number of variables is $N^3$.
- Number of clauses is polynomial in $N$, dominated by pairwise exclusivity and inequality forbidding constraints.

**Solving cost:**

- SAT is NP-complete in general, so worst-case complexity is exponential.
- In practice, modern CDCL solvers (such as Glucose3) are often very effective on structured constraints like Futoshiki.

**Space and implementation trade-offs:**

- Requires storing full CNF and SAT internal structures.
- Provides a clean and extensible formulation: adding new logical constraints usually means adding new clause templates.

**Practical behavior in this project:**

- SAT solver gives deterministic feasibility checking (SAT/UNSAT).
- Performance is strong on many instances, especially when constraints are dense enough to guide conflict learning.

## 4. Experiments and Evaluation

### 4.1. Experimental Setup

[Chèn nội dung tại đây]

### 4.2. Input and Output Test Cases

[Chèn nội dung tại đây]

### 4.3. Evaluation Metrics

[Chèn nội dung tại đây]

#### 4.3.1 Runtime

[Chèn nội dung tại đây]

#### 4.3.2 Memory Usage

[Chèn nội dung tại đây]

#### 4.3.3 Number of Inferences or Expansions

[Chèn nội dung tại đây]

### 4.4. Comparative Analysis Across Algorithms

[Chèn nội dung tại đây]

### 4.5. Self-evaluation Against Project Criteria

[Chèn nội dung tại đây]

## 5. Conclusion

This project delivers a complete and practical Futoshiki-solving system that combines multiple AI approaches with an interactive GUI. Through implementation and testing, the team verified that different paradigms (search, logic inference, and SAT) can all be applied to the same constraint problem with different trade-offs in runtime, memory, and node expansions.

From an engineering perspective, the project codebase is organized for reproducibility and extension, including standardized input/output handling, solver abstractions, and automated testing. The implementation environment is lightweight and aligned with the project requirements, using

`pygame` for GUI interaction, `python-sat` for SAT solving, and `pytest` for test execution and validation.

Overall, the project objectives were achieved: modeling constraints correctly, implementing representative AI methods, evaluating solver behavior across benchmark inputs, and presenting results through both command-line tests and GUI-based visualization.

# 6. References

[Chèn nội dung tại đây]

# A. Appendix

## A.1. Generative AI Usage Disclosure

This section documents the use of Large Language Models (LLMs) during the development of this project:

- **AI Tools Used**: Gemini, ChatGPT, Claude.
- **Usage Scenarios**:
  - Code debugging and optimization for search algorithms.
  - Formatting the LaTeX report and troubleshooting matplotlib diagrams.
  - Refining English grammar in the documentation.
- **Sample Prompts**:
  - "How should I enforce row/column uniqueness and inequality constraints efficiently in a Futoshiki backtracking solver?"
  - "In our A* Futoshiki solver, what is a good admissible heuristic to reduce node expansions on 7x7 and 9x9 inputs?"
  - "Help me format a LaTeX table that compares runtime, memory, and expanded nodes across Backtracking, Forward Chaining, Backward Chaining, SAT, and A* for the 10 input files."

## A.2. Project Deliverables

- **Demo Video**: A video demonstrating the GUI, manual gameplay, and solver animations can be found at: Click to see video.

- **Futoshiki-Puzzles Project Repository, GitHub**: Click to see repository

### A.3. Additional Tables and Charts

[Chèn nội dung tại đây]